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- Finally, consider the equation $ax = b$ where $a = b = 0$. In this case any x will satisfy the equation, i.e., there are infinitely many solutions. From these examples we conclude that a linear equation may have

  a) a unique solution, b) no solution, c) infinitely many solutions.

Introduction

- In the general form, a linear equation can be written as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = c.$$

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Example

Suppose that you are planning to cook a dish using 10 ingredients and you want it to have 2000 calories. Let a_i be the number of calories per gram of the i-th ingredient and x_i be the amount (in grams) of the i-th ingredient. Then, $a_1x_1 + a_2x_2 + \cdots + a_{10}x_{10}$ will be the calories in the recipe. Since you want to cook a dish having 2000 calories, you can express the problem by the equation

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$$a_1x_1 + a_2x_2 + \cdots + a_{10}x_{10} = 2000$$

Linear Equation Sets

- Consider a general linear system consisting of m equations in n unknowns:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

- Notice how the coefficients a_{ij} are labelled. The first index is for the row and the second index is its column. A linear equation set is said to be homogeneous if all b_i are zero. Otherwise, it is called nonhomogeneous.
- The main problem is to find a procedure for describing the solution set of a linear system. The principal procedure for solving a linear system is called 'Gaussian Reduction'.

Matrices

- To simplify the cumbersome notation for a system used above, we will introduce the notion of a matrix.

Definition

A matrix is simply a rectangular array of real numbers. An $m \times n$ matrix is an array having m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

- If $m = n$, we say A is a square matrix of dimension n. The set of all $m \times n$ matrices with real entries is denoted by $\mathbb{R}^{m \times n}$.

Matrix Addition and Vectors

Definition

The matrix sum (or simply sum) $A + B$ of two $m \times n$ matrices A and B is defined to be the $m \times n$ matrix C such that $c_{ij} = a_{ij} + b_{ij}$ for all pairs of indices (i, j). The scalar multiply αA of A by a real number α is the matrix obtained by multiplying each entry of A by α.

Example

Let

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 3 & 1 \\ 1 & 2 & 3 & 0 \end{bmatrix}.$$

Then

$$A + B = \begin{bmatrix} 2 & 2 & 3 & 2 \\ 0 & 1 & 3 & 4 \\ 1 & 2 & 4 & 5 \end{bmatrix} \text{ and } 2A = \begin{bmatrix} 2 & 0 & 0 & 4 \\ 0 & 2 & 0 & 6 \\ 0 & 0 & 2 & 10 \end{bmatrix}$$

Matrix Addition and Vectors

Definition (Zero matrix)

The $m \times n$ matrix will all components equal to zero is called the zero matrix. If 0 is the $m \times n$ zero matrix and A is any $m \times n$ matrix, then $A + 0 = A$.

A column of a matrix is usually simply called a vector. The set of all $n \times 1$ column vectors with real entries is denoted by $\mathbb{R}^n$. Vectors can be summed up or subtracted from each other in a similar way to matrix addition and subtraction. Note that the number of components in these vectors must be the same.
We can use the notation $[v_1 \;\; v_2 \;\; \cdots \;\; v_n]^T$ to express a column vector

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

Product of a matrix and a vector

Consider an $m \times n$ matrix A and a vector x with n components. Multiplying them will result in

$$Ax = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}$$

From the above expression, one can think of the representation of linear systems by matrix and vector products: $Ax = b$. In this representation, b is a column vector of m components.
Note that it is possible to multiply a matrix by a column vector if and only if the number of columns in the matrix and the number of components in the column vector are the same.

Product of two matrices

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Let the columns of B denoted by the vectors b_i, i.e., $B = [b_1 \;\; b_2 \;\; \cdots \;\; b_p]$. The matrix multiplication $C = AB$ can be expressed as

$$C = AB = [Ab_1 \quad Ab_2 \quad \cdots \quad Ab_p]$$

If we denote the (i, j)-th entry of C by c_{ij}, the matrix multiplication can be mathematically expressed as

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Note that in order to multiply two matrices, the number of columns in the first matrix and the number of rows in the second matrix must be the same.

Example

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 6 & 0 \\ -2 & 7 \end{bmatrix} = \begin{bmatrix} 1 \cdot 6 + 3 \cdot (-2) & 1 \cdot 0 + 3 \cdot 7 \\ 2 \cdot 6 + 4 \cdot (-2) & 2 \cdot 0 + 4 \cdot 7 \end{bmatrix} = \begin{bmatrix} 0 & 21 \\ 4 & 28 \end{bmatrix}$$

Product of two matrices

Example

$$\begin{bmatrix} 2 & 1 & 0 \\ -1 & 1 & 3 \\ 5 & 4 & -2 \end{bmatrix} \begin{bmatrix} -5 & -4 & 2 \\ 1 & 7 & 2 \\ 3 & 0 & -6 \end{bmatrix} = \begin{bmatrix} -9 & -1 & 6 \\ 15 & 11 & -18 \\ -27 & 8 & 30 \end{bmatrix}$$

Reduced Row Echelon Form and Row Operations

Definition

(Row echelon form) A matrix A is said to be in row echelon form if

- (i) the first non zero entry in every row is to the right of the first nonzero entry in all the rows above, and
- (ii) every entry above a first nonzero entry is zero.

The first nonzero entry in a row is called its pivot or corner entry. A matrix A in row echelon form is said to be in reduced row echelon form (or simply reduced) if every pivot entry is 1.
The reduced $n \times n$ matrix in reduced row echelon form with n pivot entries is called the $n \times n$ identity matrix. The identity matrix is denoted by I_n. For instance, $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Reduced Row Echelon Form and Row Operations

Here are some examples of reduced matrices:

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 & 0 & 9 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 3 & 0 & 9 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Reduced Row Echelon Form and Row Operations

Definition (Elementary row operations)

Let A be an arbitrary matrix. The three elementary row operations on A are as follows

- (i) interchange two rows of A,
- (ii) multiply a row of A by a nonzero scalar, and
- (iii) replace a row of A by itself plus a multiple of a different row.

Every matrix A can be put into reduced row echelon form by a sequence of elementary row operations.

Example

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

We can row reduce A as follows:

Reduced Row Echelon Form and Row Operations

Example

$$A \overset{R_2-4R_1}{\to} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix} \overset{R_3-7R_1}{\to} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \overset{-R_2/3}{\to} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -6 & -12 \end{bmatrix}$$

$$\overset{R_3+6R_2}{\to} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \overset{R_1-2R_2}{\to} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solving Linear Equations

Let A be an $m \times n$ matrix and consider the linear system $Ax = b$. Let the augmented coefficient matrix be defined as $[A \ \vdots \ b]$. Performing a sequence of row operations on the augmented coefficient matrix of a linear system gives a new system which is equivalent to the original system.

Solving Linear Equations: The Homogeneous Case

Solving a linear system $Ax = b$ involves several steps. The first step is to solve the associated homogeneous system.

A linear system $Ax = b$ is called homogeneous if $b = 0$. The efficient way to describe the solution set of a homogeneous linear system is to use vectors. The procedure of finding the solution set is as follows:

1) Find the reduced row echelon form of A, denoted by A_r.
2) For the equivalent linear system $A_r x = 0$, find the expressions of the unknown variables corresponding to the pivot elements of A_r.
3) Express the solution vector involving only free variables (non-pivot elements).
4) Express the solution vector as a linear combination of fundamental solution vectors: $x = x_1 f_1 + x_2 f_2 + \cdots + x_n f_n$.
5) Assign slack variables to the non-pivot elements and give the solution

Solving Linear Equations: The Homogeneous Case

Example

Find the solution set of the homogeneous system

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 4 & 5 \\ 2 & 4 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving Linear Equations: The Homogeneous Case

Step 1) Find the reduced row echelon for of A

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 4 & 5 \\ 2 & 4 & 0 & -2 \end{bmatrix} \overset{R_2+2R_1}{\rightarrow} \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & -2 \end{bmatrix} \overset{R_3-2R_1}{\rightarrow} \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\overset{R_1-2R_2}{\rightarrow} \begin{bmatrix} 1 & 0 & -8 & -7 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} = A_r$$

Step 2) For the equivalent linear system $A_r x = 0$, find the expressions of the unknown variables corresponding to the pivot elements of A_r.

$$\begin{bmatrix} 1 & 0 & -8 & -7 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The unknown variables corresponding to the pivot elements are x_1 and x_2:

Solving Linear Equations: The Homogeneous Case

$$x_1 = 8x_3 + 7x_4$$

$$x_2 = -4x_3 - 3x_4$$

Step 3) Express the solution vector involving only free variables (non-pivot elements).

$$x = \begin{bmatrix} 8x_3 + 7x_4 \\ -4x_3 - 3x_4 \\ x_3 \\ x_4 \end{bmatrix}.$$

Step 4) Express the solution vector as a linear combination of fundamental solution vectors: $x = x_1 f_1 + x_2 f_2 + \cdots + x_n f_n$.

$$x = \begin{bmatrix} 8x_3 + 7x_4 \\ -4x_3 - 3x_4 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \\ 1 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} 7 \\ -3 \\ 0 \\ 1 \end{bmatrix} x_4$$

Solving Linear Equations: The Homogeneous Case

Step 5) Assign slack variables to the non-pivot elements and give the solution Let $s = x_3$ and $t = x_4$. Then the solution can be expressed as

$$x = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} 8 \\ -4 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 7 \\ -3 \\ 0 \\ 1 \end{bmatrix}, \forall s, t \in \mathbb{R} \right\}$$